



DATA SCIENCE · BUSINESS INTELLIGENCE · MACHINE LEARNING

SuperStore AI Intelligence Platform

An end-to-end analytics and machine-learning platform transforming SuperStore transaction data into business insights, interactive dashboards, SQL analysis, and predictive Return_Flag classification – built across five structured Jupyter notebooks and deployed via Streamlit.

S Mohammed Kaif · Data Science · Machine Learning · AI · Data Analytics

From Raw Transactions to Predictive Intelligence

This project transforms SuperStore transaction data into an integrated analytics and predictive intelligence platform. It combines **Data Science, Business Intelligence, SQL, Machine Learning, Interactive Visualization**, and **Streamlit Deployment** – supporting both historical business understanding and forward-looking decision support.

Platform Capabilities

- Descriptive analytics & KPI intelligence
- Interactive Plotly dashboards
- Structured SQL business analysis
- Statistical & GroupBy analysis
- Binary Return_Flag classification
- Model comparison, evaluation & explainability
- Three deployed Streamlit applications

Technology Foundation

- **Python** – Core language for all data processing, analysis, and ML
- **Pandas & NumPy** – Data manipulation and numerical computation
- **Matplotlib & Seaborn** – Static exploratory visualizations
- **Plotly** – Interactive business intelligence charts
- **SQLite** – Structured SQL business analysis
- **Scikit-learn & Joblib** – ML modeling and model persistence
- **Streamlit** – Interactive web application deployment
- **Jupyter Notebook** – Iterative notebook-based development
- **ReportLab** – PDF report generation

SuperStore Transaction Data

The platform is built on two primary dataset files and one SQLite database. The final machine-learning dataset contains **5,901 rows** with **53 engineered features**, with **Return_Flag** as the binary classification target. The dataset was split into **4,720 training rows** and **1,181 testing rows**.

5,901

Total Rows

Final ML dataset

53

Features

After engineering

4,720

Training Rows

Model development

1,181

Test Rows

Model evaluation

Source Files

- `SuperStore_Sales_Dataset.csv` – Raw transaction data
- `SuperStore_Feature_Engineered.csv` – Processed dataset
- `superstore_analysis.db` – SQLite database for SQL analysis

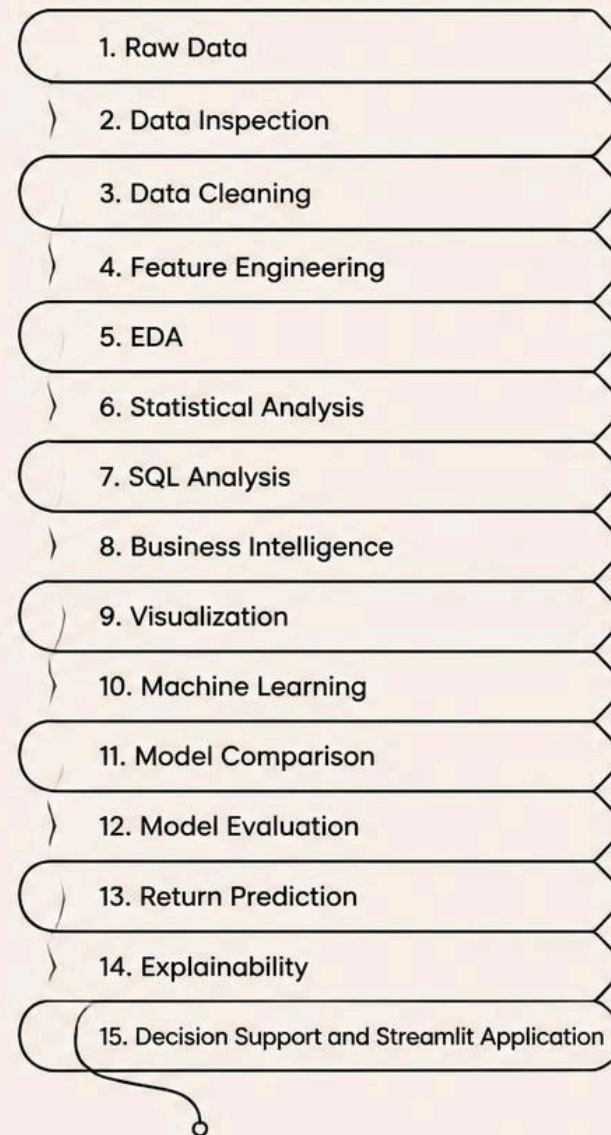
Target Variable

Return_Flag – Binary Classification

Predicts whether a transaction will result in a product return, enabling proactive return-risk management and operational planning.

End-to-End Analytical Workflow

The project follows a structured, reproducible pipeline – from raw data ingestion through to deployed business intelligence and machine-learning prediction. Each stage builds upon the previous, ensuring data quality, analytical rigor, and decision-ready outputs.



The workflow was developed across five Jupyter notebooks, each addressing a distinct analytical layer – from data preparation and EDA through to SQL analysis, interactive visualization, and complete machine-learning modeling with model comparison and evaluation.

Five-Notebook Development Process

The project was developed iteratively using Jupyter notebooks, each addressing a distinct analytical layer. This modular approach ensures reproducibility, clear separation of concerns, and a logical progression from data preparation through to predictive modeling.

01

01 — Data Inspection, Cleaning, Feature Engineering & EDA

Data loading, inspection, shape analysis, column analysis, data types, missing values, duplicates, cleaning, feature engineering, univariate, bivariate, and exploratory data analysis.

02

02 — GroupBy & Statistical Business Analysis

Pandas GroupBy, aggregation, sales analysis, profit analysis, category, regional, segment, and product analysis, statistical analysis, and business performance evaluation.

03

03 — SQL for Data Analysis

SQLite queries using SELECT, WHERE, GROUP BY, ORDER BY, HAVING, CASE, aggregate functions, subqueries, and business KPI extraction from `superstore_analysis.db`.

04

04 — Plotly Business Insights

Interactive visualizations covering sales trends, profit trends, category performance, regional performance, segment analysis, product performance, and profitability comparisons.

05

05 — ML Return Prediction & Modeling

Complete ML workflow: data preparation, target definition, feature preparation, train/test split, preprocessing, model training, comparison, evaluation, selection, prediction, and model export.

SuperStore Key Performance Indicators

The platform generates a comprehensive set of business KPIs from SuperStore transaction data. These metrics form the foundation of all downstream business intelligence, trend analysis, and strategic decision support.

\$1.57M

Total Sales

\$1,565,804.32 across all orders and categories

\$175K

Total Profit

\$175,262.11 net profit generated

11.19%

Profit Margin

Net profit as a percentage of total sales

3,003

Total Orders

Across 773 unique customers

22,317

Quantity Sold

Total units sold across all product categories

\$521.41

Avg Order Value

With \$58.36 profit generated per order

These KPIs are computed through Pandas aggregation, GroupBy analysis, and SQL queries – and are surfaced interactively across all three Streamlit applications for real-time stakeholder access.

Four Dimensions of Business Intelligence

The platform analyzes SuperStore data across four intelligence dimensions – sales, profitability, customer behavior, and time trends – each explored through interactive Plotly visualizations, SQL queries, and statistical analysis.



Sales Intelligence

Sales performance across categories, regions, segments, and products. Time-based trend analysis reveals yearly and monthly sales patterns. SQL queries identify top-performing categories and regional sales contribution.



Profitability Intelligence

Profit performance, margin analysis, and profitability breakdowns by category, region, and product. Identifies high-margin opportunities and areas requiring strategic attention.



Customer Intelligence

Analysis of 773 customers, their contribution to revenue, Average Order Value (\$521.41), and Profit Per Order (\$58.36). Segment-level analysis reveals behavioral patterns across customer groups.



Time Intelligence

Yearly and monthly trends for sales, profit, and margin. Time-series visualizations support seasonal pattern identification and forward-looking performance planning.

Return_Flag Prediction Model

The platform applies binary classification to predict **Return_Flag**. Multiple algorithms were trained and compared, with **F1 Score** as the primary selection metric – balancing Precision and Recall for reliable positive-case identification.

Models Evaluated

- Logistic Regression
- Random Forest
- Extra Trees
- Gradient Boosting
- Hist Gradient Boosting
- KNN
- SVM (when available)
- XGBoost (when available)

Selected Model: Logistic Regression

Model Performance — Test Set (1,181 rows)

- Accuracy: 1.0000
- Precision: 1.0000
- Recall: 1.0000
- F1 Score: 1.0000
- ROC-AUC: 1.0000

 **Model Validation Notice:** Perfect test-set performance should be thoroughly investigated before production deployment.

Recommended validation areas include: target leakage, target-derived features, duplicate or highly related observations, dataset construction characteristics, and validation using genuinely unseen future data. This is a validation consideration – not a confirmed issue.

Understanding Evaluation Metrics

Five complementary metrics assess classifier performance. **F1 Score** is the primary selection criterion because Return_Flag prediction requires balancing reliable positive identification with minimizing missed returns.

Accuracy

Overall classification correctness – proportion of correct predictions across both classes.

Precision

Reliability of positive predictions – of transactions flagged as returns, how many actually were.

Recall

Ability to identify positive cases – of actual returns, how many were correctly flagged.

F1 Score

Harmonic mean of Precision and Recall.
Primary metric for model selection, balancing false positives and false negatives.

ROC-AUC

Ability to distinguish between return and non-return transactions across all classification thresholds.

Structured Business Analysis with SQLite

SQLite was used to perform structured, query-driven business analysis via `superstore_analysis.db`. SQL enables reproducible, auditable KPI extraction and segment-level investigation – complementing Pandas and GroupBy workflows.

SQL Techniques Applied

- **SELECT & WHERE** – Targeted data retrieval and filtering
- **GROUP BY & HAVING** – Aggregation by category, region, and segment
- **ORDER BY** – Ranked performance comparisons
- **CASE** – Conditional logic for custom business metrics
- **Aggregate Functions** – SUM, COUNT, AVG for KPI computation
- **Subqueries** – Nested analytical queries

Analytical Questions Addressed

- Which categories generate the most sales?
- Which regions generate the highest profit?
- Which segments contribute the most revenue?
- Which products have strong or weak profitability?
- What are the major business KPIs?